

GABRIEL NILSSON

Cambridge, UK · oagn2@cam.ac.uk · [linkedin.com/in/gab-nil](https://www.linkedin.com/in/gab-nil) · github.com/g-nilsson

EDUCATION

University of Cambridge

Cambridge, United Kingdom

MPhil in Scientific Computing – High Performance Computing

2025-2026

- Saven Cambridge Scholarship – merit-based full scholarship, £30,000 award
- Thesis: Coherent Mesh Representation of Unstructured Polyhedral Meshes

Minerva University

San Francisco, Seoul, Taipei, Hyderabad, Buenos Aires, London, and Berlin

Bachelor of Science in Computational Sciences (Mathematical and Computational Modeling)

2021-2025

- GPA: 3.93 / 4.00. Major GPA: 3.97 / 4.00
- World's most innovative University [2022](#), [2023](#), [2024](#), & [2025](#) (WURI)
- Very selective global program focused on interdisciplinarity; studying in 7 countries with highly international cohort

Swedish Naval Academy

Karlskrona, Sweden

System Technician Onboard Submarines

2020-2021

- 10 of ~5,000 were selected
 - Soft skills: Military discipline, leadership, working in teams, and managing classified information
 - Hard skills: Network protocols, security, radar, sonar, GPS, and advanced electronics
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RESEARCH

- MPhil thesis: “Coherent Mesh Representation of Unstructured Polyhedral Meshes” – supervised by Hrvoje Jasak
 - Median ~40x speedup of large-scale CFD simulations, relative to current methods.
 - Implementing the Weiß et al. (2024) coherent mesh format in OpenFOAM, enabling parallel I/O and domain decomposition in CFD simulations.
 - Bachelor's thesis: “Spatial discretization and its impact on physical simulations” – supervised by Lucas Tambasco
 - Implemented a 3D Dielectric Breakdown model, isolating cell connectivity's effect on simulation outcomes. Showing that grid decision meaningfully impacts results, which previous research has assumed it does not
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EXPERIENCE

Minerva University

Global Rotation

Lead Peer Tutor, Teaching Assistant, and Curriculum Intern

Aug 2021 – Apr 2025

- Teaching Assistant for Differential Equations, Probability & Statistics, and Calculus
- Designed the University's AI literacy class and developed grading dashboard for faculty
- Tutored Python programming and math to 100+ students after placing first in freshman coding assessment
 - Graded 400+ assignments, provided in-depth feedback, and designed the programming curriculum

Game Development Projects

Borås, Sweden

Lead programmer (3 years)

Aug 2017 – May 2020

- Was Programming Lead and Scrum Master for 3 different year-long projects of 8+ people
 - Qualified for a national competition and won prizes every year, with 60+ teams participating each year
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ACCOMPLISHMENTS

- “Best Beginner Hack” at CalHacks, the **world's largest collegiate hackathon**, 370+ participants (2021)
 - **40th best coder in Sweden**, national high-school programming olympiad, 250+ participants (2019)
 - “Best Game” in Swedish game development competition, 60+ teams (2020)
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SKILLS

- **Skills:** *Advanced:* Tutoring, mathematics, statistical analysis, and high-performance computing. *Intermediate:* ML
- **Languages/Software:** *Advanced:* Python (numpy, scipy, pytorch, pandas) [10 years]; C#, Unity, Git [8 years]; Linux [3 years]; C++ [2 years]